ACADEMIC PROJECT ABSTRACT

WATCHIFY

YOUR TIME , YOUR STYLE

PROJECT NAME	WATCHIFY – WATCH COLLECTION MANAGEMENT SYSTEM
CLIENT-SIDE	HTML5, CSS3, JavaScript
TECHNOLOGY	PHP
DATABASE	MySQL
SUBMITTED BY	BRIJESH RAMANI
ACADEMIC YEAR	2026-27

1. Project Abstract

Watchify is a dynamic, database-driven web application designed to showcase, manage, and sell a curated collection of wristwatches online. Built using PHP for server-side logic and MySQL for persistent data storage, the system bridges the gap between watch retailers and customers by offering a seamless digital shopping experience.

The platform enables customers to browse watches by brand and category, view detailed product information, add items to a shopping cart, and place orders securely. On the administrative side, the system provides a dedicated dashboard for managing the product catalogue, tracking orders, and maintaining customer records — all through an intuitive, role-based interface.

This project demonstrates the practical application of core web development concepts including client-server architecture, relational database design, CRUD operations, session handling, and form validation, making it a comprehensive academic showcase of full-stack web development using the PHP–MySQL stack.

2. Description of the Website

Purpose

Watchify aims to digitize the traditional watch-retail experience by providing an organized, searchable, and visually appealing online catalogue of watches, along with a streamlined ordering and administrative workflow.

Key Modules

- User Module – Registration, login, profile management, and secure authentication.
- Catalogue Module – Browse watches by category/brand, view specifications, pricing, and stock availability.
- Cart & Order Module – Add to cart, update quantities, checkout, and track order status.
- Admin Module – Add, update, or remove watches; manage categories; view and process customer orders.
- Reporting Module – Generate sales and inventory reports for business insights.

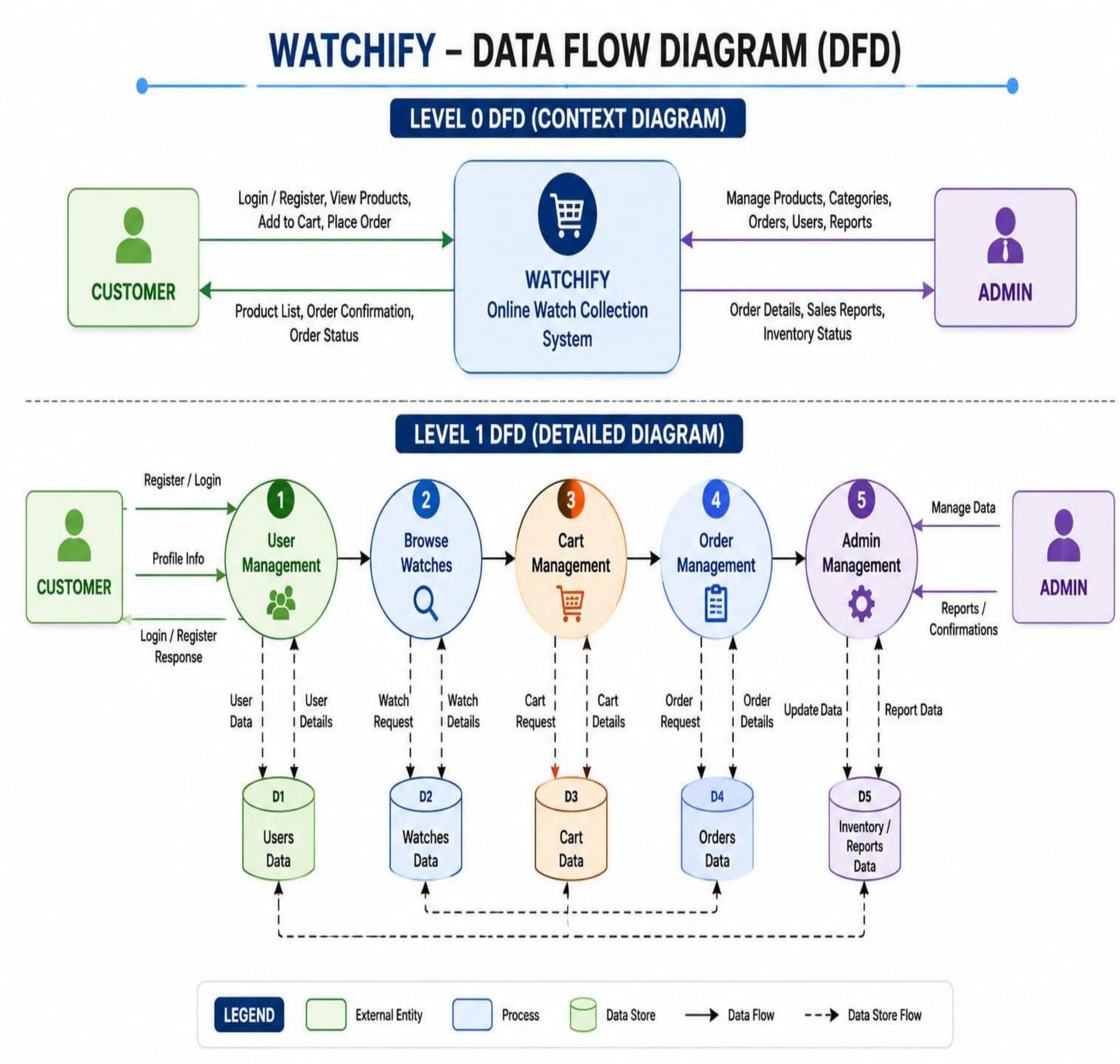
3. Folder Structure

The project follows a modular folder structure that separates administrative, user-facing, and core functionalities, ensuring maintainability and scalability.

```
Watchify/
├── admin/
│   ├── dashboard.php
│   ├── manage_watches.php
│   ├── manage_category.php
│   ├── view_orders.php
│   └── reports.php
├── user/
│   ├── register.php
│   ├── login.php
│   ├── profile.php
│   ├── cart.php
│   └── order.php
├── includes/
│   ├── db_connect.php
│   ├── header.php
│   ├── footer.php
│   └── functions.php
├── assets/
│   ├── css/style.css
│   ├── js/script.js
│   └── images/  (watch images & icons)
├── config/
│   └── config.php
├── index.php
├── product_details.php
├── checkout.php
├── logout.php
└── database/
    └── watchify.sql
```

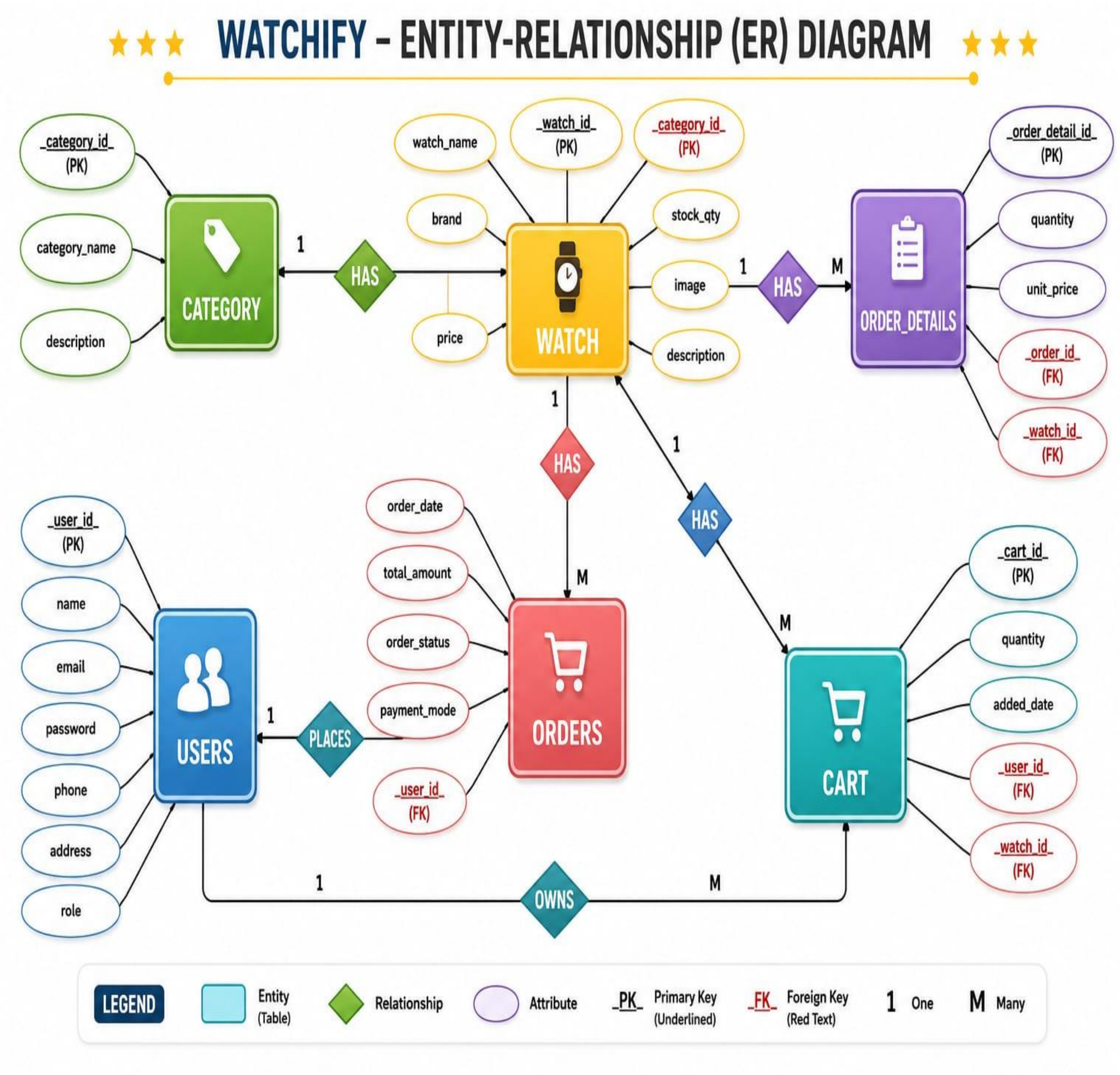
4. Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how information moves between the Customer, Admin, system processes, and the underlying data stores within Watchify.



5. Entity-Relationship (ER) Diagram

The ER Diagram below depicts the relational database design of Watchify, highlighting Primary Keys (PK), Foreign Keys (FK), and the relationships (1:M) between entities.



6. Data Dictionary

The data dictionary defines the structure of each table used in the Watchify database, specifying field names, data types, sizes, and key constraints (Primary Key / Foreign Key).

Table: tbl_users

Field Name	Data Type	Size	Key	Description
user_id	INT	11	**PK**	Unique identifier for each user (Auto Increment)
name	VARCHAR	100	-	Full name of the user
email	VARCHAR	100	-	Unique email address used for login
password	VARCHAR	255	-	Encrypted user password
phone	VARCHAR	15	-	Contact number of the user
address	TEXT	-	-	Delivery / billing address
role	VARCHAR	20	-	Defines user type: Customer or Admin

Table: tbl_category

Field Name	Data Type	Size	Key	Description
category_id	INT	11	**PK**	Unique identifier for each category (Auto Increment)
category_name	VARCHAR	100	-	Name of the watch category (e.g. Analog, Smart)
description	TEXT	-	-	Brief description of the category

Table: tbl_watch

Field Name	Data Type	Size	Key	Description
watch_id	INT	11	PK	Unique identifier for each watch (Auto Increment)
watch_name	VARCHAR	150	-	Name/model of the watch
brand	VARCHAR	100	-	Brand or manufacturer name
category_id	INT	11	FK	References category_id in tbl_category
price	DECIMAL	10,2	-	Selling price of the watch
stock_qty	INT	11	-	Available stock quantity
image	VARCHAR	255	-	File path of the watch image
description	TEXT	-	-	Detailed product description

Table: tbl_orders

Field Name	Data Type	Size	Key	Description
order_id	INT	11	PK	Unique identifier for each order (Auto Increment)
user_id	INT	11	FK	References user_id in tbl_users
order_date	DATETIME	-	-	Date and time the order was placed
total_amount	DECIMAL	10,2	-	Total value of the order
order_status	VARCHAR	30	-	Pending / Shipped / Delivered / Cancelled
payment_mode	VARCHAR	30	-	Mode of payment (COD, Card, UPI, etc.)

Table: tbl_order_details

Field Name	Data Type	Size	Key	Description
order_detail_id	INT	11	PK	Unique identifier for each order line (Auto Increment)
order_id	INT	11	FK	References order_id in tbl_orders
watch_id	INT	11	FK	References watch_id in tbl_watch
quantity	INT	11	-	Quantity of the watch ordered
unit_price	DECIMAL	10,2	-	Price per unit at the time of order

Table: tbl_cart

Field Name	Data Type	Size	Key	Description
cart_id	INT	11	**PK**	Unique identifier for each cart entry (Auto Increment)
user_id	INT	11	**FK**	References user_id in tbl_users
watch_id	INT	11	**FK**	References watch_id in tbl_watch
quantity	INT	11	-	Quantity added to the cart
added_date	DATETIME	-	-	Date and time item was added to cart

7. Conclusion

Watchify successfully demonstrates a complete e-commerce workflow for a watch collection business, integrating PHP and MySQL to deliver a functional, user-friendly, and scalable web application. The structured database design, clear data flow, and modular folder organization reflect sound software engineering practices suitable for academic evaluation and real-world extension.